Vansh Gupta

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CodeChef 4-Star (max 1856) | Codeforces Pupil (max 1340) | Bhopal, India

Professional Summary

Third-year B.Tech student at IIIT Bhopal (CPI 8.49/10) specializing in **large language models, vision-language models, and multimodal generative systems**. Currently ML Intern at Moleculyst designing **multimodal Video LLM inference pipelines** with structured evaluation loops for production deployment. Shipped LLM-driven systems across the full research-to-production loop: a **Reinforcement Learning environment for LLaMA 3.1** (Triage AI, 0.71/1.0 reward), an autonomous **5-agent LangGraph pipeline** using the Gemini API (Equitas AI), and a **multimodal agentic VLM workspace** with automated critique loops (Moleculyst Studio). Strong foundations in probability, linear algebra, and statistical modeling; actively deepening into evaluation methodology, prompt engineering, and quantitative research. Eager to apply LLM and multimodal generative ML techniques to ambitious drug-discovery problems.

Education

IIIT Bhopal Bhopal, Madhya Pradesh, India
B.Tech in Information Technology (CS & Engineering Curriculum) | CPI: 8.49 / 10 Aug 2024 – May 2028

- **Relevant Coursework**: Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra, Probability & Statistics, Discrete Mathematics, Operating Systems, Database Management Systems, Computer Networks

Technical Skills

LLM / VLM Stack: LLaMA 3.1, Gemini API, Vision-Language Models, Multimodal LLMs, Transformers, RAG, Prompt Engineering, Structured Evaluation, OpenEnv

Agentic Frameworks: LangChain, LangGraph, Multi-Agent Orchestration, Tool Use, Automated Critique Loops, Reinforcement Learning Environments

ML / Deep Learning: PyTorch, TensorFlow, Scikit-learn, Generative Models, LSTM, XGBoost, SHAP

Research & Math Foundations: Probability Theory, Linear Algebra, Statistical Modeling, Mathematical Optimization

Languages: Python, C++, C, SQL

Engineering & Infrastructure: FastAPI, Docker, Git/GitHub, REST API Design, Pydantic, Software Testing

Cloud & Deployment: AWS, GCP (Cloud Run, Vertex AI), Scalable Pipeline Architecture, Observability, Monitoring

Competitive Programming (C++): CodeChef 4-Star (max 1856), Codeforces Pupil (max 1340) – DP, graphs, segment trees, combinatorics

Experience

ML Intern Feb 2026 – Present
Moleculyst Remote

- Designing and shipping **multimodal Video LLM inference pipelines** in Python – integrating vision-language architectures, preprocessing stages, and structured evaluation loops – delivering Moleculyst's core inference system for structured understanding across complex operational video datasets.
- Built reproducible **evaluation frameworks for LLM outputs** that standardized quality assessment across model versions, enabling consistent benchmarking and faster iteration on model and prompt changes – without manual review.
- Expanding scope into **applied quantitative research** – probability theory, mathematical optimization, statistical modeling – to inform model selection, evaluation design, and uncertainty quantification across generative ML experiments.
- Collaborating with cross-functional stakeholders to capture user requirements, iterate on technical designs, and deliver software on schedule in a cooperative team environment.

Projects

Triage AI – Reinforcement Learning Environment for LLaMA 3.1 *Python, FastAPI, OpenEnv, Pydantic, Docker, LLaMA 3.1* 2025

- Architected a **reinforcement learning environment** for automated email classification on **LLaMA 3.1 (8B)**, implementing structured OpenEnv state-machine cycles (reset / step / observe) with **multi-level prompt-engineering chains**, achieving **0.71 / 1.0 reward accuracy**.
- Containerized the full system with Docker and enforced API contracts with Pydantic – reproducible, environment-isolated deployment that can be reliably reset and re-evaluated across test cycles, supporting rigorous LLM benchmarking workflows.

Moleculyst Studio – Multimodal Agentic VLM Workspace *Python, Gemini API, VLMs, OpenCV, Docker* 2025

- Built a multi-step **agentic editing pipeline** composing **VLM-based visual grounding**, targeted edit operators, and an **automated critique loop** – replacing manual step-by-step review with a single-command end-to-end

multimodal workflow.

- Streamed structured reasoning traces, tool calls, and per-step quality scores into a unified monitoring view – enabling engineers to inspect, debug, and intervene at any stage of the agentic pipeline.

Equitas AI – Multi-Agent LLM Pipeline for Bias Mitigation *Python, LangGraph, FastAPI, Gemini API, Vertex AI, GCP* 2025

- Designed a **5-agent autonomous LangGraph pipeline** for dataset bias auditing, orchestrating iterative mitigation loops via the **Gemini API** that converged at Disparate Impact Ratio (DIR) ≥ 0.80 – the U.S. legal fairness threshold – across all evaluated datasets.
- Integrated **SHAP matrices with Gemini-driven counterfactual narration**, generating row-level natural-language bias reasoning through an interactive review interface usable by non-technical stakeholders.
- Deployed a keyless FastAPI backend on Google Cloud Run via Vertex AI – fully serverless, autoscaling, no stored secrets.

PharmAI – Pharmaceutical Batch Quality Forecasting *Python, LSTM, XGBoost, SHAP, Scikit-learn* 2025

- Engineered an LSTM + XGBoost ensemble for pharmaceutical batch quality prediction on historical process data, achieving **>90% accuracy** on held-out validation batches – directly applicable to drug-manufacturing quality workflows.
- Implemented SHAP-driven anomaly attribution surfacing top contributing process variables per flagged batch for interpretable corrective action.

Achievements

- **Meta × PyTorch OpenEnv Hackathon – Grand Finalist** (Bangalore, 2025): Top-tier ranking out of 52,000+ global participants for shipping a production-ready RL environment for LLMs.
- **CodeBidz Hackathon – 1st Place**: Won as a solo developer, executing the full product lifecycle under competition time constraints.
- **OpenAI × Outskill AI Builders Hackathon**: Shortlisted in Top 1,000 developers nationwide to ship a production-grade live AI product.
- **JEE Main 2024 – 98.16 percentile**: Top 1% nationally out of 1.2M+ candidates; qualified for JEE Advanced.
- **SRMC Mathematics Competition – AIR 27 (Finals)**: National finalist in the tournament jointly organized by IIT Bombay & IIT Madras.